



Benchmarking AI-generated thin-slice CT under clinical reconstruction conditions: a multicohort study



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Thin-slice CT provides finer anatomical depiction, but in some clinical settings, only thick-slice CT images are available, resulting in reduced visible anatomical detail. Simply resampling to a smaller through-plane spacing cannot alleviate partial-volume-related image degradation, motivating deep learning-based CT slice synthesis (CSS) as an auxiliary strategy. Here, we release Multi-ratio Slice Synthesis CT (MSS-CT), a multi-ratio real-paired dataset comprising thick-slice and 1 mm CT reconstructed from identical raw acquisitions, covering 3 mm-to-1 mm (500 pairs) and 5 mm-to-1 mm (1500 pairs) synthesis. Using MSS-CT, we developed BasicCSS as a reference model, evaluated it against representative CSS methods in a multicohort setting, and compared real-paired with pseudo-paired training. BasicCSS achieved the highest performance in internal and external testing, whereas pseudo-paired training led to marked declines across all evaluated methods, consistent with mismatch between simulated thick-slice inputs and clinically reconstructed thick-slice CT. Downstream consistency analyses showed that BasicCSS-synthesized 1 mm CT had higher consistency with real 1 mm CT-derived segmentation and radiomic outputs than conventional interpolation. In a two-radiologist reader study, synthesized 1 mm CT reviewed alongside original 5 mm CT as an auxiliary view received high usefulness and confidence-support scores. These results suggest that AI-generated thin-slice CT may serve as an auxiliary image when real thin-slice CT is unavailable, and that MSS-CT and BasicCSS enable reproducible benchmarking for future CSS development and evaluation under clinically reconstruction conditions.

Computed tomography (CT) is a core diagnostic imaging modality, and its clinical and quantitative value depends in part on how volumetric information is reconstructed and represented^{1,2}. During CT reconstruction, raw projection data are reconstructed into a series of two-dimensional slices according to predefined parameters, such as slice thickness, slice interval, and reconstruction kernel, yielding a three-dimensional volumetric representation. Among these parameters, slice thickness is a key determinant of spatial resolution and image fidelity. Thin-slice CT provides denser axial

sampling, improving through-plane resolution and anatomical delineation^{3–5}. In routine practice, however, the availability and use of thin-slice CT may be limited by image-noise and dose trade-offs, storage burden, and health-system specific cost considerations^{6–8}. Thick-slice CT therefore remains widely used, despite reduced through-plane resolution and blurring from partial-volume effects, in which mixed tissue components within a voxel reduce image contrast and structural sharpness⁹. These slice-thickness-related effects can influence visual image interpretation and

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quantitative CT measurements^{10,11}. Enhancing thick-slice CT toward thin-slice CT quality may therefore improve volumetric representation and support subsequent image-based interpretation and quantitative analysis.

Conventional interpolation is commonly used to resample thick-slice CT to a thinner slice spacing, but it standardizes image sampling without alleviating partial-volume-related image degradation caused by thick-slice reconstruction. To address this limitation, deep learning-based CT slice synthesis (CSS) has been developed to generate thin-slice CT from thick-slice inputs when thin-slice reconstruction is desired but unavailable^{12–21}. Existing CSS methods can be broadly grouped into reformatted-view super-resolution^{12,15,21}, volume super-resolution^{13,14}, and slice-wise interpolation paradigms^{16–20}, which differ in how they model through-plane information. Although recent CSS methods have shown promise, their development or evaluation has often relied on thick-slice inputs simulated by artificially degrading thin-slice CT^{19–21}. Such pseudo-paired data provide a practical surrogate when real-paired reconstructions are unavailable, but they only approximate the reconstruction differences observed between real thick-slice and thin-slice CT series. It therefore remains uncertain whether CSS models trained on pseudo-paired data can preserve information needed for image interpretation and quantitative analysis when applied to real thick-slice inputs.

This uncertainty reflects a central data limitation in current CSS research, as large-scale real-paired datasets with thick-slice and thin-slice CT series reconstructed from identical raw acquisitions remain limited. Without such data, evaluation may be shaped by simplified degradation assumptions rather than by scanner-specific reconstruction characteristics, slice-thickness-related blurring, and other differences observed in clinical CT reconstruction. A publicly available real-paired dataset has provided an important step toward realistic CSS evaluation¹⁶, but its limited scale and single synthesis ratio restrict systematic assessment across different synthesis difficulties. A larger multi-ratio real-paired dataset is therefore needed as a common data foundation for developing CSS methods and evaluating their performance under realistic reconstruction conditions.

With such a data foundation, CSS evaluation can extend beyond visual performance to whether synthesized thin-slice CT supports downstream quantitative image analysis and clinical image review. For quantitative image analysis, the value of synthetic thin-slice CT depends on whether it can produce outputs that are more consistent with those derived from real thin-slice CT^{22–24}. In anatomy-based quantitative assessment, ready-to-use segmentation tools increasingly provide anatomical measurements and downstream quantitative outputs from CT images^{25–27}. Variations in slice thickness, however, can affect segmentation performance and reduce the consistency of segmentation-derived measurements^{28–30}. In feature-based quantitative assessment, radiomics provides another common pathway for extracting CT information as quantitative features describing intensity and texture patterns³¹. Radiomics pipelines commonly use interpolation-based resampling to standardize voxel spacing³², but this step does not resolve the information loss caused by thick-slice reconstruction^{33–35}. Alongside these quantitative endpoints, synthesized images should also be evaluated for whether they provide information that supports human interpretation rather than visual realism alone³⁶. For synthesized thin-slice CT, its practical value may depend on whether it can support CT image interpretation when only thick-slice CT is available.

To address this data gap and enable CSS evaluation beyond image similarity alone, we release Multi-ratio Slice Synthesis CT (MSS-CT), a large-scale, multi-ratio, real-paired chest CT dataset. MSS-CT includes 2000 subjects and supports 3 mm to 1 mm and 5 mm to 1 mm synthesis using thick-slice and thin-slice series reconstructed from identical raw acquisitions (Fig. 1a). We further developed BasicCSS as a reference model and benchmarked it against representative CSS methods and conventional interpolation across internal and external real-paired datasets (Fig. 1b). The evaluation assessed synthesis fidelity, the effect of real-paired versus pseudo-paired training, downstream structural and radiomic consistency, and the perceived value of synthesized 1 mm CT as an auxiliary view (Fig. 1c). By making MSS-CT and BasicCSS publicly available, this study further

supports reproducible CSS development and evaluation under clinically reconstruction conditions.

Results

Study cohorts

MSS-CT was constructed as a real-paired chest CT dataset for CT slice synthesis, with paired thick-slice and 1 mm CT series reconstructed from identical raw acquisitions. As shown in Table 1, MSS-CT included 2000 subjects across two internal subsets. MSS-CT 3 mm supported the 3 mm to 1 mm synthesis task, and MSS-CT 5 mm supported the 5 mm to 1 mm synthesis task (Fig. 1a). MSS-CT was used for model development and internal evaluation, whereas two independent real-paired datasets, RPLHR-CT¹⁶ and EC-CT, were used for external testing of the 5 mm to 1 mm task (Fig. 1b). Across all paired datasets, the thicker-slice CT series was used as input, and the matched real 1 mm CT series was used as the reference.

Internal CT slice synthesis performance on MSS-CT

We compared BasicCSS with 16 representative methods across the three main CSS paradigms under identical real-paired training and testing settings. These comparators included reformatted-view super-resolution^{12,15,21}, volume super-resolution^{13,14}, slice-wise interpolation^{16–20}, and representative natural image super-resolution and video frame interpolation methods adapted to CSS^{37–42}. In the main text, ACVTT²¹, ResVox¹³, and CTHNet¹⁸ are reported as representative comparators for the three paradigms, with complete quantitative results for all 16 methods provided in Supplementary Tables S1 and S2. Synthesis performance was evaluated against the matched real 1 mm CT reference using PSNR and SSIM measured in the axial, coronal, and sagittal planes.

BasicCSS achieved the highest PSNR and SSIM in all three planes for the 3 mm to 1 mm synthesis task (Fig. 2a). Trilinear interpolation showed lower performance across PSNR and all three SSIM metrics, with a PSNR of 37.97 ± 0.93 dB and axial, coronal, and sagittal SSIM values of 0.967 ± 0.005 , 0.964 ± 0.006 , and 0.963 ± 0.006 , respectively. All comparisons between BasicCSS and the evaluated methods were statistically significant for each metric, with all $p < 0.001$ (Supplementary Table S1).

For the 5 mm to 1 mm synthesis task, BasicCSS again achieved the highest PSNR and three-plane SSIM (Fig. 2b). BasicCSS reached a PSNR of 44.09 ± 1.43 dB, compared with 42.11 to 43.48 dB for the other CSS methods. The performance gap between methods was larger in the higher-ratio 5 mm to 1 mm task, particularly relative to trilinear interpolation. All comparisons between BasicCSS and the evaluated methods were statistically significant for each metric, with all $p < 0.001$ (Supplementary Table S2).

Qualitative comparisons supported the quantitative findings. In selected examples, CSS methods reduced residual blurring relative to trilinear interpolation, although remaining fine-structure loss was more evident in the 5 mm to 1 mm task than in the 3 mm to 1 mm task. High-contrast structures such as bones were preserved by most CSS methods. Among the compared methods, BasicCSS more consistently preserved fine airway walls and small vessels and showed less residual blurring, particularly in the 5 mm to 1 mm task (Fig. 2c, d).

Ablation experiments supported the contributions of the block-wise decoder and cross-view refinement module (Supplementary Table S3 and Supplementary Fig. S1). Each component improved synthesis performance and convergence, and the full BasicCSS model outperformed an enlarged block-wise decoder variant without cross-view refinement, suggesting that the gain was not explained by increased model capacity alone.

External CT slice synthesis performance on real-paired datasets

External testing for the 5 mm to 1 mm synthesis task was performed on two independent real-paired datasets, RPLHR-CT and EC-CT, to assess generalization beyond the internal MSS-CT data. The main text reports the same representative comparators as in the internal evaluation, and complete quantitative results for all 16 methods are provided in Supplementary Tables S4 and S5.

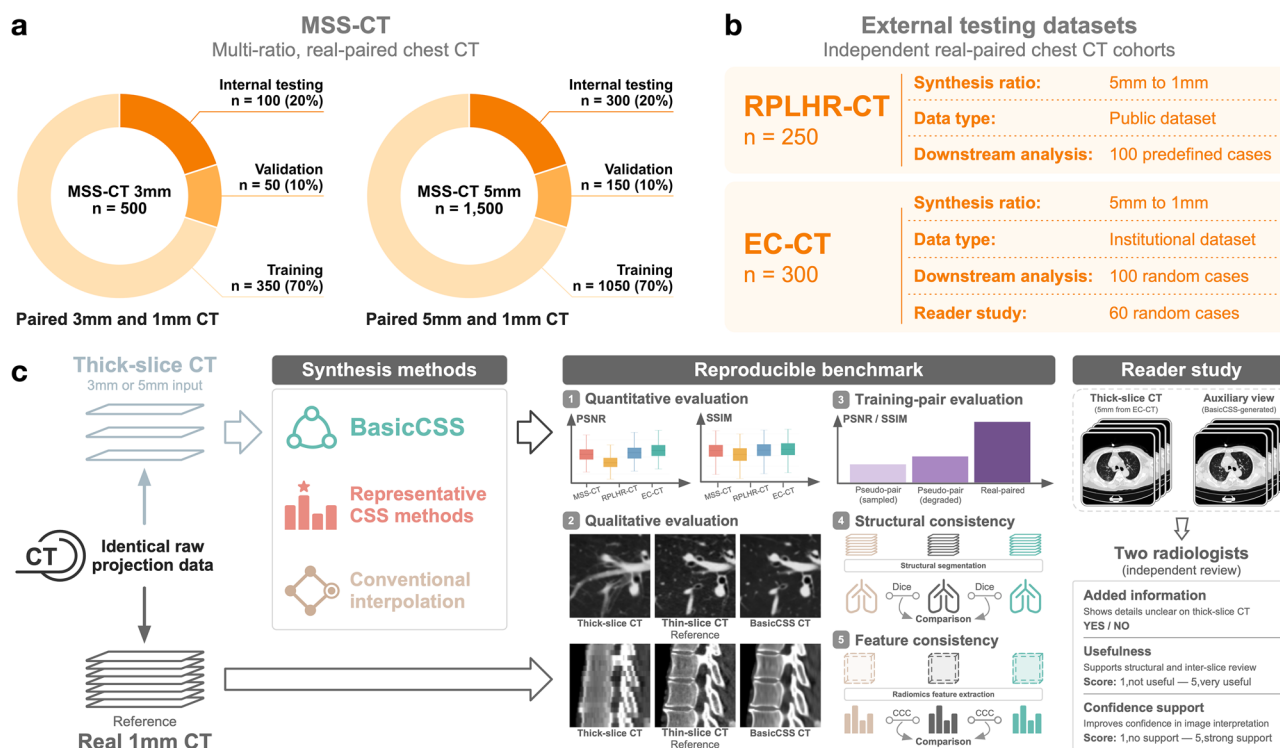


Fig. 1 | Study overview. **a** Composition of MSS-CT, a multi-ratio real-paired CT dataset. MSS-CT included two subsets. MSS-CT 3 mm was used for the 3 mm to 1 mm synthesis task and included 500 subjects, of whom 350 were assigned to training, 50 to validation, and 100 to internal testing. MSS-CT 5 mm was used for the 5 mm to 1 mm synthesis task and included 1,500 subjects, of whom 1,050 were assigned to training, 150 to validation, and 300 to internal testing. **b** External testing datasets for the 5 mm to 1 mm synthesis task. RPLHR-CT was a public real-paired CT dataset with 250 subjects, with 100 predefined cases used for downstream analyses. EC-CT was an institutional real-paired CT dataset with 300 subjects, with 100 randomly selected cases used for downstream analyses and 60 randomly selected cases used for the reader study. **c** Evaluation framework. Thick-slice CT was used as

input for conventional interpolation, representative CSS methods, and BasicCSS, and the matched real 1 mm CT reconstructed from identical raw projection data served as the reference. The reproducible benchmark included quantitative and qualitative evaluation, training-pair evaluation comparing pseudo-paired and real-paired training, structural consistency analysis, and radiomic feature consistency analysis. The reader study evaluated BasicCSS-generated 1 mm CT as an auxiliary view to original 5 mm thick-slice CT, using added information, usefulness, and confidence support as assessment items. CCC concordance correlation coefficient, CSS CT slice synthesis, PSNR peak signal-to-noise ratio, SSIM structural similarity index measure.

On both RPLHR-CT and EC-CT, BasicCSS achieved the highest PSNR and three-plane SSIM among the evaluated methods, whereas trilinear interpolation showed lower performance across all metrics. The differences between BasicCSS and the evaluated methods were statistically significant for each metric in both external test sets (all $p < 0.001$; Fig. 3a, b and Supplementary Tables S4 and S5). Absolute metric values differed between the two datasets, with lower values on RPLHR-CT than on EC-CT, suggesting dataset-level differences in synthesis difficulty.

Qualitative examples from RPLHR-CT and EC-CT are shown in Fig. 3c, d. In both external datasets, BasicCSS-generated thin-slice CT more consistently preserved fine structures and showed less residual blurring than trilinear interpolation and the representative CSS methods. Although quantitative metrics were lower on RPLHR-CT than on EC-CT, this dataset-level difference was less apparent in the selected qualitative examples.

We performed an external-center adaptation analysis to assess whether MSS-CT-trained BasicCSS could support site-specific fine-tuning. On both RPLHR-CT and EC-CT, MSS-CT-pretrained fine-tuning achieved the highest PSNR and three-plane SSIM among the evaluated adaptation settings. These results suggest that MSS-CT-trained BasicCSS can provide a useful initialization for adapting CSS models to external-center data, with complete results provided in Supplementary Table S6.

Effect of real-paired and pseudo-paired training

To quantify the effect of training-pair realism on CSS performance and method ranking, we compared real-paired training with two pseudo-paired

strategies. In the sampled pseudo-paired setting, thin-slice images were sampled at axial positions corresponding to the target thick-slice series. In the degraded pseudo-paired setting, thin-slice CT volumes were artificially degraded along the axial direction to simulate thick-slice inputs^{19–21}. The analysis included BasicCSS, ACVTT, ResVox, CTHNet, and two recent pseudo-paired methods, I3Net¹⁹ and SFCLI-Net²⁰. All models were trained under the three training-pair settings and evaluated on real-paired MSS-CT test sets. Real-paired training produced higher PSNR than both pseudo-paired settings for all evaluated methods in both synthesis tasks, with a larger effect in the higher-ratio 5 mm to 1 mm task (Fig. 4 and Supplementary Table S7).

Degraded pseudo-paired training generally outperformed sampled pseudo-paired training, but both remained below real-paired training. In both synthesis tasks, ACVTT achieved the highest PSNR under pseudo-paired settings, whereas BasicCSS ranked highest under real-paired training, indicating that pseudo-paired training can change conclusions about relative method performance. The same pattern was observed for three-plane SSIM (Supplementary Table S8).

Segmentation-derived structural consistency

Segmentation-derived structural consistency was assessed by comparing segmentations from conventional interpolation-resampled 1 mm CT (Tri-CT) and BasicCSS-generated 1 mm CT (CSS-CT) with reference segmentations from matched real 1 mm CT (Fig. 5a). This analysis included the MSS-CT 3 mm and MSS-CT 5 mm internal test sets, the predefined 100-case RPLHR-CT test set¹⁶, and 100 randomly selected EC-CT cases. Ten

Table 1 | Study datasets and their use in evaluation

	MSS-CT dataset		External datasets	
	MSS-CT 3 mm	MSS-CT 5 mm	RPLHR-CT	EC-CT
Synthesis task	3 mm to 1 mm	5 mm to 1 mm	5 mm to 1 mm	5 mm to 1 mm
Participants	500	1500	250	300
Age (y)	64.5 [52.0, 71.0]	34.0 [24.0, 61.0]	NA	25.0 [22.0, 29.0]
Male/Female (%)	45.6/54.4	57.7/42.3	NA	67.6/32.4
Data split				
Training	350	1,050	Not used	Not used
Validation	50	150	Not used	Not used
Testing	100	300	250	300
Clinical status				
Healthy	0	1,000	0	200
Abnormal	0	500	0	100
Unknown	500	0	250	0
CT manufacturer	Philips	Neusoft	Philips	Philips
In-plane resolution (mm)	0.564–0.977	0.449–0.924	0.604–0.795	0.477–0.977
Tube voltage (kVp)	100–140	120	NA	100
Tube current (mA)	74–801	165–755	NA	184–808
Analysis cases				
Structural consistency	100	300	100	100
Radiomic consistency	100	300	100	100
Reader study	Not used	Not used	Not used	60
Data availability	Publicly released	Publicly released	Publicly available	Limited access

Values are shown as numbers of participants unless otherwise indicated. Age is shown as median [IQR]. Male and female values are shown as percentages. Analysis cases indicate the number of cases included in each downstream evaluation. All paired thick-slice and 1 mm CT series were reconstructed from the same raw acquisitions. NA not available, IQR interquartile range.

thoracic structures were evaluated using Dice similarity coefficients, including the heart, aorta, lung vessels, trachea, T3 and T7 vertebrae, bilateral fifth ribs, and bilateral seventh ribs. To characterize the external subsets used for downstream consistency analyses, PSNR and three-plane SSIM were also calculated, with complete results provided in Supplementary Table S9.

Across all datasets and anatomical structures, CSS-CT showed higher consistency with real 1 mm CT-derived segmentations than Tri-CT (all $p < 0.01$; Fig. 5b and Supplementary Table S10). The improvement from Tri-CT to CSS-CT was more pronounced in the 5 mm to 1 mm task than in the 3 mm to 1 mm task. With Tri-CT, the largest decreases were observed for lung vessels, trachea, and bone structures. CSS-CT reduced this task-related difference, with similar mean Dice values in MSS-CT 3 mm and MSS-CT 5 mm.

Among the 5 mm to 1 mm datasets, MSS-CT 5 mm and EC-CT showed similar overall CSS-CT consistency, although their structure-specific patterns differed. CSS-CT showed higher consistency in MSS-CT 5 mm for lung vessels and trachea, whereas EC-CT showed higher consistency for most bone structures. Heart and aortic consistency remained high in both datasets. The two external datasets also showed different structure-specific patterns. EC-CT generally showed higher consistency than RPLHR-CT for the heart and bone structures, whereas aortic consistency was high in both datasets. RPLHR-CT showed lower Tri-CT consistency for bone structures but larger gains after CSS-CT, and the trachea showed a larger increase from Tri-CT to CSS-CT in RPLHR-CT than in EC-CT.

Radiomic feature consistency

Radiomic feature consistency was evaluated using matched real 1 mm CT-derived features as the reference (Fig. 6a). Lung ROIs defined on real 1 mm CT were mapped to thick-slice CT, Tri-CT, and CSS-CT, and radiomic

features extracted from each image type were compared using the concordance correlation coefficient (CCC). This analysis included the complete MSS-CT 3 mm and MSS-CT 5 mm internal test sets, the predefined 100-case RPLHR-CT test set¹⁶, and 100 randomly selected EC-CT cases, with results summarized by feature group (Fig. 6b, c and Supplementary Table S11).

Across all four datasets, CSS-CT showed higher radiomic feature consistency with real 1 mm CT-derived features than thick-slice CT and Tri-CT, as reflected by CCC distributions and reproducible feature proportions. Tri-CT showed a different pattern, with lower overall mean CCC than thick-slice CT but a higher proportion of reproducible features. This apparent discrepancy may reflect the feature-dependent and nonuniform effect of trilinear interpolation. Compared with thick-slice CT, Tri-CT tended to show a wider CCC range, suggesting that interpolation changed radiomic features in inconsistent directions and to different extents. Consequently, some features exceeded the reproducibility threshold of 0.85, whereas reduced concordance in other features lowered the overall mean CCC.

Radiomic feature consistency differed across feature groups. The advantage of CSS-CT over thick-slice CT and Tri-CT was most evident for LoG features, whereas wavelet features had lower CCC values and fewer reproducible features across datasets, suggesting greater sensitivity to dataset and reconstruction differences. Radiomic consistency also varied by synthesis ratio and dataset. MSS-CT 3 mm showed higher consistency than MSS-CT 5 mm, consistent with the lower synthesis difficulty of the 3 mm to 1 mm task. Among the 5 mm to 1 mm datasets, EC-CT showed a profile closer to MSS-CT 5 mm, whereas RPLHR-CT had lower absolute radiomic consistency, particularly for wavelet features. These findings suggest that radiomic consistency after CSS depends on both synthesis ratio and dataset-specific reconstruction characteristics.

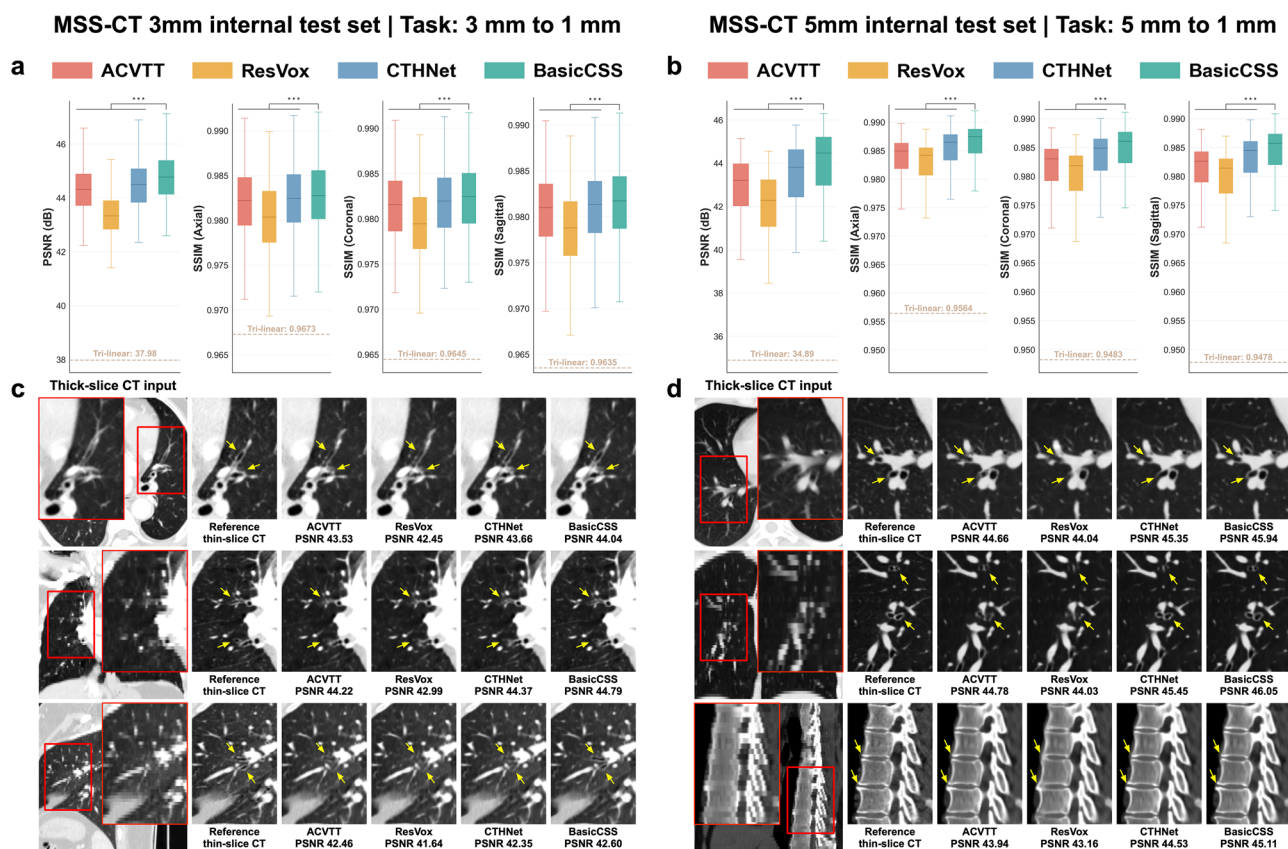


Fig. 2 | Internal CT slice synthesis performance on MSS-CT. Quantitative performance of representative CT slice synthesis methods on the MSS-CT 3 mm internal test set for the 3 mm to 1 mm task (a) and the MSS-CT 5 mm internal test set for the 5 mm to 1 mm task (b). Performance was measured using PSNR and SSIM in the axial, coronal, and sagittal planes. Box plots show the distribution of case-level results, with the center line indicating the median and the box indicating the interquartile range. Dashed horizontal lines indicate median values of trilinear interpolation. Statistical comparisons were performed using two-sided paired Wilcoxon signed-rank tests, with Bonferroni correction for multiple comparisons,

and are shown above the box plots ($p < 0.001$ as ***). Representative qualitative comparisons on the MSS-CT 3 mm internal test set (c) and MSS-CT 5 mm internal test set (d). For each example, the thick-slice CT input, reference thin-slice CT, and images synthesized by representative methods are shown. The examples are shown from top to bottom in axial, coronal, and sagittal views, respectively, to illustrate both in-plane and through-plane image appearance. Red boxes indicate the enlarged regions, and yellow arrows indicate regions with visible differences among methods. PSNR values are shown below each synthesized image. PSNR, peak signal-to-noise ratio; SSIM, structural similarity index measure.

Reader study

The reader study assessed whether BasicCSS-synthesized 1 mm CT could provide added value when reviewed with original 5 mm CT as an auxiliary view. Sixty EC-CT cases, including 30 healthy cases and 30 pneumonia cases, were independently reviewed by two radiologists. For each case, original 5 mm CT and synthesized 1 mm CT were displayed side by side without access to real 1 mm CT. Readers were blinded to health status. Guided by CT image-quality criteria that emphasize diagnostic information and visible anatomical detail⁴³, the readers assessed added information, usefulness, and confidence support. Added information was rated as yes or no, and usefulness and confidence support were scored on 5-point scales. Usefulness and confidence-support ratings were high and similar between healthy and pneumonia cases (Fig. 7 and Table 2). Across 120 reader-case ratings, the mean usefulness score was 4.83, and the mean confidence-support score was 4.69. No reader assigned scores of 1 to 3 for either item.

Discussion

In this study, we introduced MSS-CT, a multi-ratio real-paired chest CT dataset of 2000 subjects for CT slice synthesis, and used it to evaluate AI-generated thin-slice CT under clinical reconstruction conditions. We developed BasicCSS as a reference model and benchmarked it against representative CSS methods and conventional interpolation across internal and external real-paired datasets. BasicCSS achieved higher synthesis performance, and the comparison between real-paired and pseudo-paired

training showed the importance of clinically matched reconstruction pairs for CSS development and evaluation. In downstream analyses, BasicCSS-generated 1 mm CT showed higher consistency with real 1 mm CT-derived structural and radiomic outputs than conventional interpolation. The reader study further suggested that synthesized 1 mm CT may provide a useful auxiliary view when real thin-slice CT is unavailable.

A central motivation for MSS-CT was to bring CSS development and evaluation closer to the input conditions expected in clinical use. Pseudo-paired training remains attractive because it can convert existing thin-slice CT into surrogate paired data, but its validity depends on how closely the simulated thick-slice input approximates clinically reconstructed thick-slice CT. Real thick-slice CT reflects through-plane signal aggregation, partial-volume blurring, and scanner-specific reconstruction characteristics, which are only partly captured by slice sampling or interpolation-based degradation. In our experiments, pseudo-paired training produced lower performance than real-paired training, with a larger decrease in the sampled pseudo-paired setting than in the degraded pseudo-paired setting. These results suggest that real-paired data remain important for verifying synthesis performance and method improvements. The flexibility of pseudo-paired training also allows studies to use different source datasets and pseudo-pair generation strategies, which may complicate method comparison and limit consensus on relative CSS performance. MSS-CT was therefore released as a large-scale real-paired dataset for CSS development, evaluation, and controlled comparison. The previously released RPLHR-CT dataset

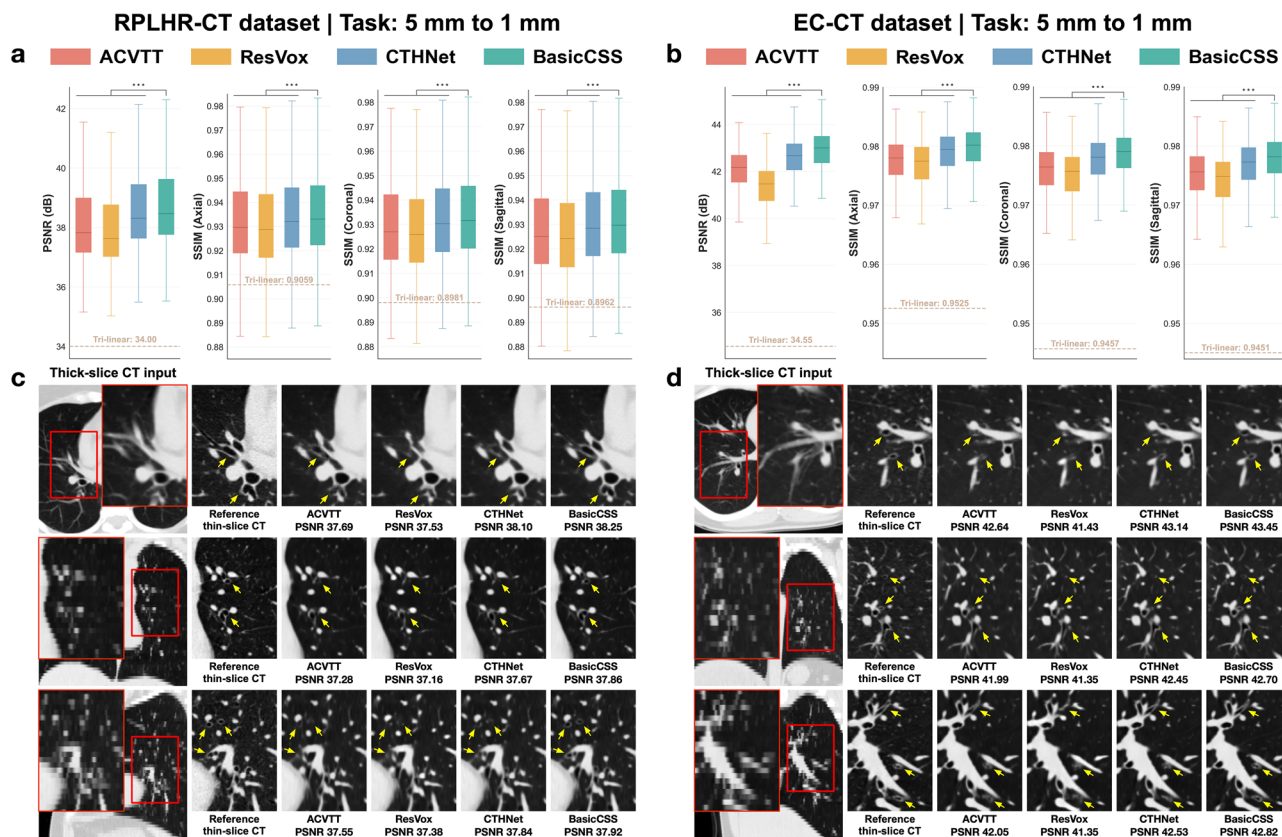


Fig. 3 | CT slice synthesis performance for the 5mm to 1mm synthesis task on external testing datasets. Quantitative performance of representative CT slice synthesis methods on the RPLHR-CT dataset (a) and the EC-CT dataset (b). Performance was measured using PSNR and SSIM in the axial, coronal, and sagittal planes. Box plots show the distribution of case-level results, with the center line indicating the median and the box indicating the interquartile range. Dashed horizontal lines indicate median values of trilinear interpolation. Statistical comparisons were performed using two-sided paired Wilcoxon signed-rank tests, with Bonferroni correction for multiple comparisons, and are shown above the box plots ($p < 0.001$ as

***). Representative qualitative comparisons on the RPLHR-CT dataset (c) and the EC-CT dataset (d). For each example, the thick-slice CT input, reference thin-slice CT, and images synthesized by representative methods are shown. The examples are shown from top to bottom in axial, coronal, and sagittal views, respectively, to illustrate both in-plane and through-plane image appearance. Red boxes indicate the enlarged regions, and yellow arrows indicate regions with visible differences among methods. PSNR values are shown below each synthesized image. PSNR peak signal-to-noise ratio; SSIM structural similarity index measure.

represented an important step toward real-paired CSS research¹⁶, but its modest scale and single synthesis ratio limited systematic assessment across different synthesis difficulties. MSS-CT complements this resource by providing a larger multi-ratio data foundation. Its use alongside RPLHR-CT as an independent external dataset also showed that real-paired datasets can still differ in reconstruction characteristics and image distributions, making cross-dataset generalization an important direction for future research.

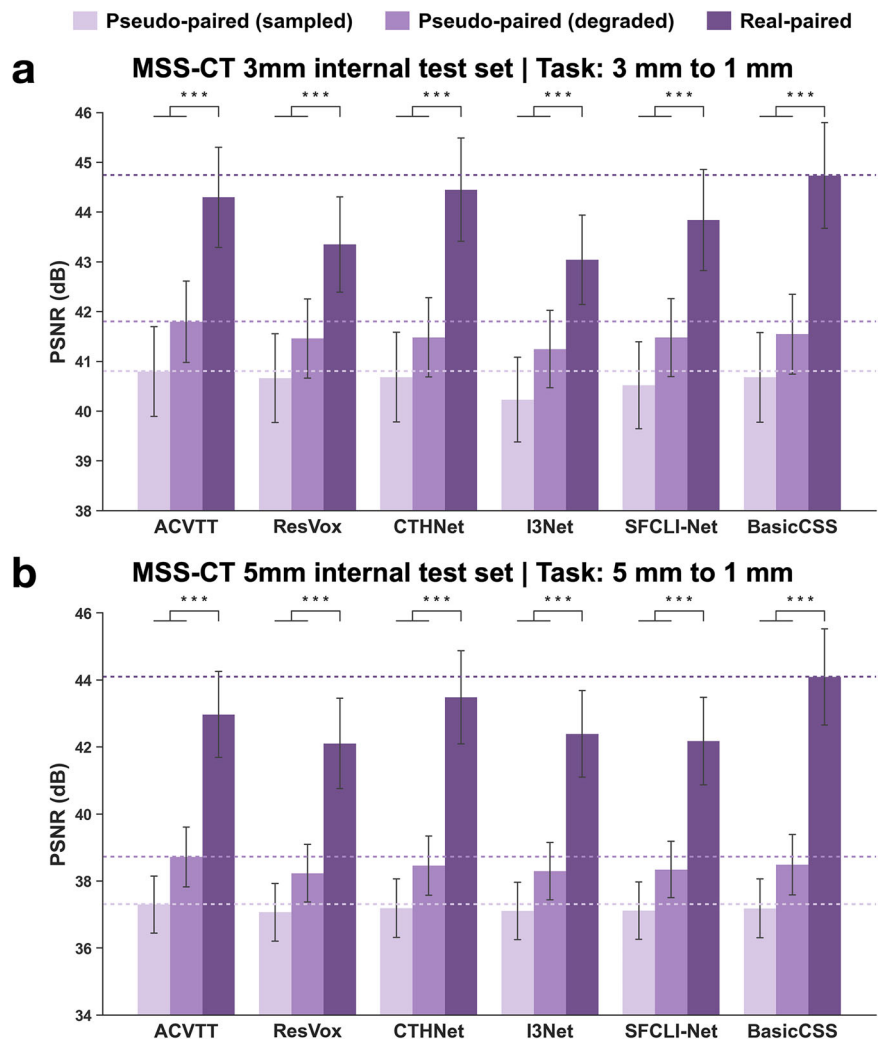
The MSS-CT benchmark showed lower performance across methods in the 5 mm to 1 mm task than in the 3 mm to 1 mm task, consistent with the greater through-plane spacing and stronger partial-volume averaging in 5 mm inputs. Conventional interpolation performed worse because resampling can standardize slice spacing but cannot recover information lost during thick-slice reconstruction. Across both synthesis settings, slice-wise interpolation methods generally achieved higher synthesis performance than reformatted-view super-resolution and volume super-resolution methods. This pattern is consistent with the CSS task, in which missing intermediate slices are mainly inferred from the local relationship between adjacent thick-slice inputs. These observations suggest that explicit modeling of axial relationships remains important for CSS under real-paired reconstruction conditions.

BasicCSS achieved consistently higher synthesis performance than the comparator methods on the MSS-CT internal test sets. Relative to the strongest comparator, CTHNet, BasicCSS improved PSNR by 0.29 dB in the 3 mm to 1 mm task and by 0.61 dB in the 5 mm to 1 mm task. These

improvements were larger than the corresponding PSNR gaps between CTHNet and the next-ranked methods. Visual comparisons showed less residual blurring and more stable preservation of fine airway walls and small vessels, particularly in the more challenging 5 mm to 1 mm task. Ablation results supported the contributions of the block-wise decoder and cross-view refinement module. These findings are consistent with the design rationale of BasicCSS. In thick-slice CT, each reconstructed slice reflects through-plane signal aggregation and partial-volume blurring, and the missing intermediate 1 mm slices are mainly constrained by the local anatomy between adjacent thick-slice positions. The block-wise decoder was designed to refine these local axial regions, making the synthesis task more localized and tractable. The cross-view refinement module was applied after axial feature expansion and block-wise decoding, when the feature representation is closer to the target thin-slice space than to the original thick-slice input. This allows sagittal and coronal continuity to be refined using reconstruction-ready axial features, which is particularly relevant for structures affected by through-plane blurring, such as small vessels and airway walls.

External testing supported the generalizability of BasicCSS under independent real-paired conditions, as it maintained the highest PSNR and three-plane SSIM on both RPLHR-CT and EC-CT. Absolute performance differed between the two external datasets, with lower image-similarity metrics on RPLHR-CT than on EC-CT. This difference should be interpreted in the context of dataset-specific image acquisition, reconstruction

Fig. 4 | Effect of real-paired and pseudo-paired training on CT slice synthesis performance. PSNR of representative CT slice synthesis methods trained under sampled pseudo-paired, degraded pseudo-paired, or real-paired settings. Results are shown for the MSS-CT 3 mm internal test set for the 3 mm to 1 mm task (a) and the MSS-CT 5 mm internal test set for the 5 mm to 1 mm task (b). All methods were evaluated on real-paired test sets. Bars indicate mean values, and error bars indicate standard deviations. Color-matched dashed horizontal lines indicate the highest PSNR achieved within each training-pair setting in each panel. For each method, statistical comparisons were performed between real-paired training and each pseudo-paired training setting using two-sided paired Wilcoxon signed-rank tests with Bonferroni correction ($p < 0.001$ as ***). PSNR peak signal-to-noise ratio.



protocols, and noise characteristics rather than as limited external generalization alone. BasicCSS also achieved performance on the predefined RPLHR-CT test subset that was comparable to the original RPLHR-CT study despite being evaluated without fine-tuning. In visual review, RPLHR-CT appeared to have higher noise than MSS-CT and EC-CT, but radiation dose information was not available, so this observation could not be directly attributed to dose differences. These findings support the need for complementary assessments beyond pixel-wise similarity metrics.

Prior studies have shown that slice thickness can affect segmentation-derived measurements^{28–30} and radiomic features^{33–35}. CSS has also been explored as a way to improve radiomic feature reproducibility³³ and support automated CT image analysis^{18,22,23}. Our study extends this evidence by evaluating whether synthetic thin-slice CT generated from real thick-slice inputs can produce structural and radiomic outputs that are more consistent with those derived from real 1 mm CT. Across multicohort datasets, BasicCSS-generated 1 mm CT showed higher consistency with real 1 mm CT-derived structural and radiomic outputs than conventional interpolation. The structural analysis suggested greater benefit for structures affected by partial-volume-related degradation, although small tubular structures such as lung vessels remained difficult to recover completely. In the radiomic analysis, BasicCSS improved feature consistency, with stronger preservation of LoG features than wavelet features. This pattern may indicate that CSS may better restore local density transitions and edge-related structures than multiscale texture patterns, which remain more sensitive to reconstruction differences. These findings indicate that downstream quantitative consistency is an informative endpoint for CSS evaluation, while the remaining

gaps in small structures and reconstruction-sensitive features identify targets for future method improvement.

Previous CSS and virtual thin-slice CT studies have used reader-based assessments to evaluate image quality, anatomical visibility, or diagnostic performance^{14,18,44}. Our reader study instead focused on an auxiliary-view setting, assessing whether synthesized 1 mm CT could provide useful additional information when reviewed alongside the original 5 mm CT. This setting reflects a practical scenario in which real thin-slice CT is unavailable, but an additional synthesized thin-slice view may help readers review image information affected by partial-volume averaging. BasicCSS-generated 1 mm CT was perceived as useful and confidence-supporting by both radiologists, including in cases with visible lung abnormalities. Because BasicCSS operates on reconstructed CT images, the synthesized 1 mm series can be generated after routine CT reconstruction and reviewed with the original 5 mm CT as an auxiliary image. In an inference-time test, BasicCSS completed synthesis within 1 min per CT volume on a single NVIDIA Quadro RTX 4000 GPU with 8 GB VRAM, suggesting that its computational requirements are feasible for potential clinical use.

Several limitations should be acknowledged. First, the benchmark was limited to chest CT, and external validation was performed only for the 5 mm to 1 mm task. Further validation is needed in other anatomical regions and in external datasets for the 3 mm to 1 mm setting. Second, although RPLHR-CT and EC-CT provided independent real-paired testing, metadata were incomplete, particularly for RPLHR-CT. Missing radiation dose, reconstruction kernel, and acquisition parameters limited our ability to explain performance differences between external datasets. Third, the

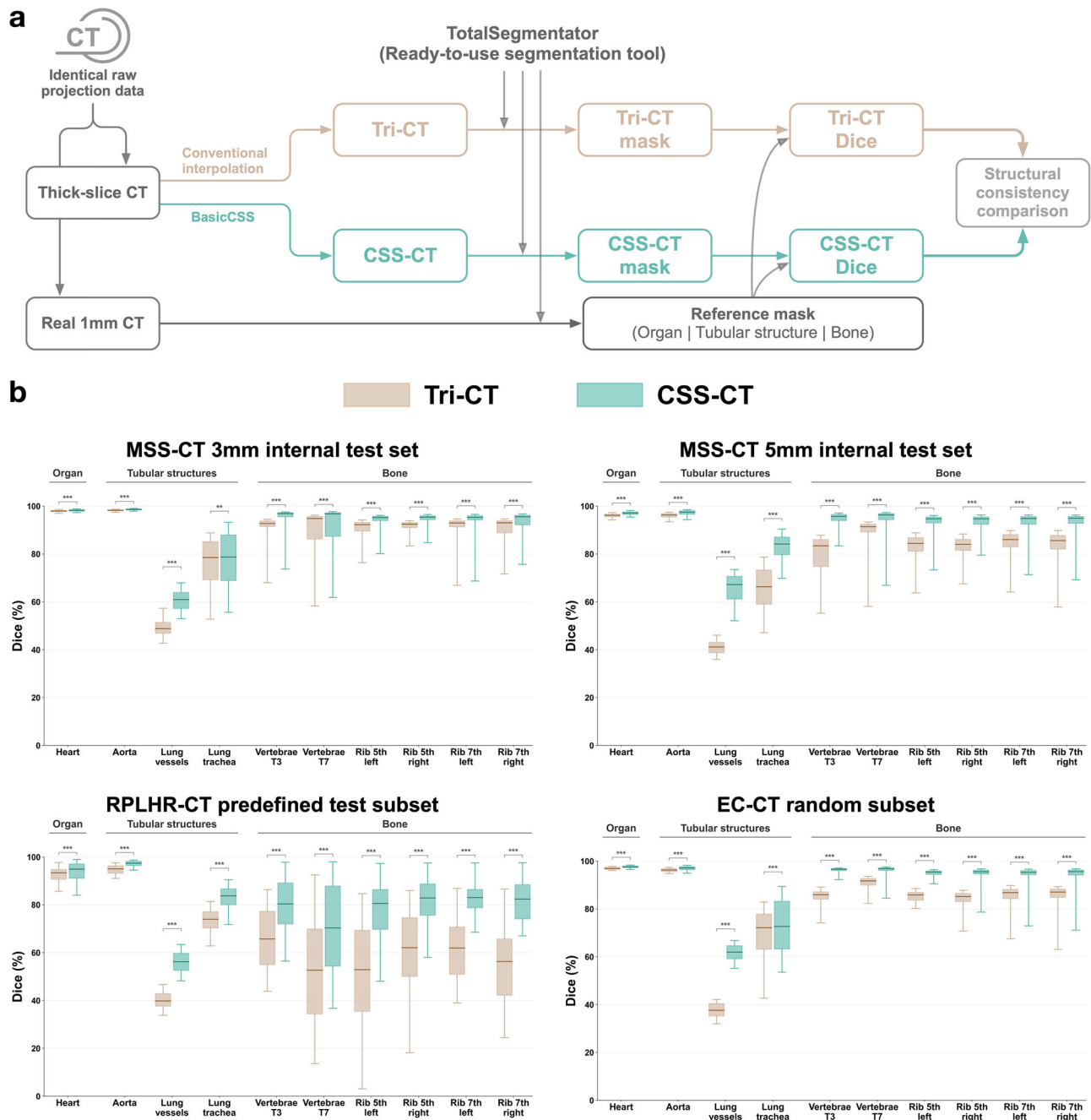


Fig. 5 | Structural consistency with real 1 mm CT-derived segmentations.

a Workflow for segmentation-derived structural consistency assessment. TotalSegmentator was applied to matched real 1 mm CT, Tri-CT, and CSS-CT. Segmentation masks derived from real 1 mm CT were used as the reference, and Dice similarity coefficients were calculated to compare Tri-CT- and CSS-CT-derived masks with the reference masks. **b** Segmentation-derived structural consistency across four real-paired datasets. Dice similarity coefficients are shown for the MSS-CT 3 mm internal test set, the MSS-CT 5 mm internal test set, the predefined 100-case RPLHR-CT test set, and 100 randomly selected EC-CT cases. Evaluated

structures were grouped into organ, tubular structure, and bone categories. Dice values are displayed as percentages, with higher values indicating greater consistency with real 1 mm CT-derived segmentations. Box plots show case-level Dice distributions, with the center line indicating the median and the box indicating the interquartile range. Statistical comparisons between Tri-CT and CSS-CT were performed using two-sided paired Wilcoxon signed-rank tests with Bonferroni correction ($p < 0.01$ as **, and $p < 0.001$ as ***). CSS-CT Basic CSS-generated CT, Tri-CT trilinear interpolation-resampled CT.

downstream analyses assessed consistency with real 1 mm CT-derived outputs rather than task-specific clinical performance. Structural consistency was evaluated using automated segmentation outputs, and radiomic consistency was assessed using randomly sampled lung ROIs. Further validation with manual annotations and lesion- or anatomy-specific radiomic analyses is needed. Fourth, the reader study did not include a diagnostic-accuracy comparison or an interpolation-resampled

comparator. Because readers were aware that the auxiliary series was synthesized, the high usefulness and confidence-support scores may partly reflect expectation or context effects. Future blinded reader studies with diagnostic endpoints are needed to determine whether the perceived auxiliary value translates into measurable interpretive benefit.

Future work should extend this framework to other settings in which thick-slice reconstructed CT remains clinically relevant, such as abdominal

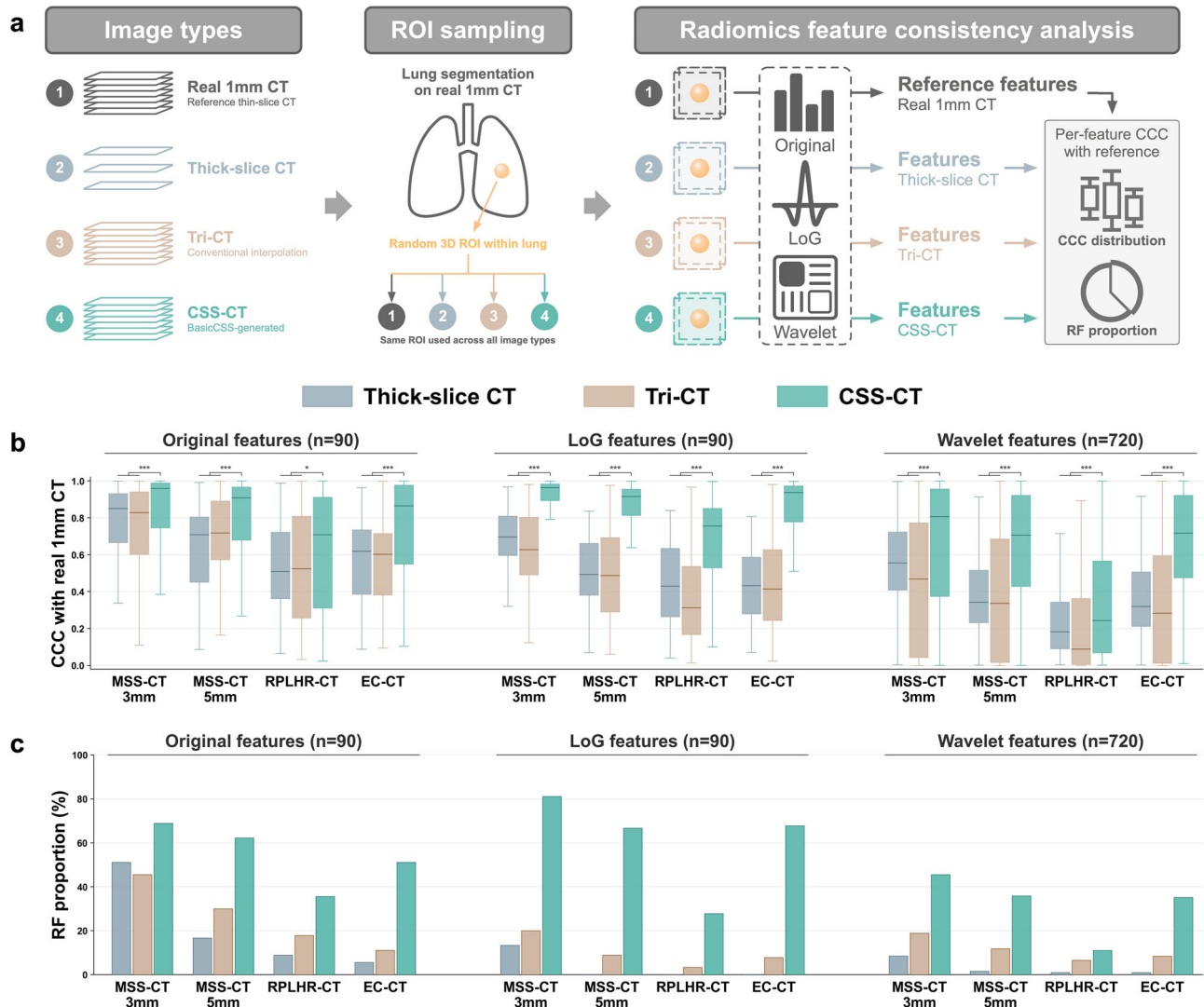


Fig. 6 | Radiomic feature consistency with real 1mm CT-derived features.
a Workflow for radiomic feature consistency assessment. Lung segmentation was performed on real 1 mm CT. For each case, a random 3D ROI was sampled within the segmented lung region, and the same ROI location was used across matched thick-slice CT, Tri-CT, and CSS-CT. Radiomic features were extracted from each image type and compared with real 1 mm CT-derived features using the concordance correlation coefficient. **b** CCC distributions across radiomic feature groups. CCC values relative to real 1 mm CT-derived features are shown for original, LoG, and wavelet features in the MSS-CT 3 mm internal test set, MSS-CT 5 mm internal test set, predefined 100-case RPLHR-CT test set, and 100 randomly selected EC-CT cases. Box plots show feature-level CCC distributions, with the center line

indicating the median and the box indicating the interquartile range. Statistical comparisons were performed using two-sided paired Wilcoxon signed-rank tests with Bonferroni correction, with significance shown above the box plots ($p < 0.05$ as *, $p < 0.01$ as **, and $p < 0.001$ as ***). **c** Proportion of reproducible radiomic features across datasets and feature groups. Reproducible features, defined as features with $\text{CCC} \geq 0.85$ relative to real 1 mm CT-derived features, are shown for original, LoG, and wavelet features in thick-slice CT, Tri-CT, and CSS-CT. Tri-CT trilinear interpolation-resampled CT, CSS-CT BasicCSS-generated CT, ROI region of interest, LoG Laplacian of Gaussian, CCC concordance correlation coefficient.

and head CT. Larger reader studies should focus on subtle findings and diagnostically challenging cases where thin-slice information may provide added value. Small-sample fine-tuning and test-time adaptation also warrant further study, as they may help adapt CSS methods to site-specific scanner, protocol, and reconstruction characteristics when only limited paired data are available.

In conclusion, MSS-CT provides a large-scale, multi-ratio, real-paired dataset for reproducible CT slice synthesis evaluation under clinically reconstruction conditions. BasicCSS serves as a reference model and showed consistent performance across internal and external real-paired testing, while the pseudo-paired comparison highlighted the importance of clinically matched reconstruction pairs. The downstream structural, radiomic, and reader-based evaluations further suggest that AI-generated 1 mm CT may serve as an auxiliary image for

quantitative assessment and CT image review when real thin-slice CT is unavailable. The released dataset, reference model, and evaluation design provide a reusable foundation for future CSS development and benchmarking.

Methods

Ethical approval

This study was approved by the institutional review boards of the General Hospital of Ningxia Medical University (KYLL-2025-1831), the Second Affiliated Hospital of Shenyang Medical College (2025-SYEYLL-034), and the Chinese PLA General Hospital (S2023-498-01). Because this retrospective study used de-identified data, the requirement for informed consent was waived by the institutional review boards. All institutional data were de-identified before analysis. The publicly available RPLHR-CT

dataset contained de-identified data and was used in accordance with its data-use terms.

Dataset collection

MSS-CT was constructed from two institutional chest CT cohorts. The MSS-CT 3 mm subset comprised 500 paired 3 mm and 1 mm chest CT examinations collected from the General Hospital of Ningxia Medical University between November 2023 and December 2024. The MSS-CT 5 mm subset comprised 1500 paired 5 mm and 1 mm chest CT examinations collected from the Second Affiliated Hospital of Shenyang Medical College between March 2023 and June 2025, including 1000 healthy subjects and 500 patients with abnormal chest CT findings. For both subsets, paired thick-slice and 1 mm CT series were reconstructed from the same raw

projection data. The thick-slice series was used as the model input, and the paired 1 mm series was used as the reference. The MSS-CT 3 mm subset was randomly divided into training, validation, and internal test sets, whereas the MSS-CT 5 mm subset was divided using stratified random sampling by health status.

Two external real-paired datasets were used to evaluate the 5 mm to 1 mm synthesis task. RPLHR-CT¹⁶ was used as a public external dataset, and EC-CT was used as an institutional external dataset from Chinese PLA General Hospital. All cases in both external datasets were used for external synthesis testing. The predefined 100-case RPLHR-CT test set and 100 randomly selected EC-CT cases were used for downstream structural and radiomic consistency analyses. Among the selected EC-CT cases, 60 cases, including 30 healthy subjects and 30 patients with pneumonia, were used for the reader study.

Paired thick-slice and 1 mm CT series were aligned according to DICOM slice positions. Because the paired series were reconstructed from the same raw projection data, affine or deformable registration was not applied. For all paired CT series, the slice interval was identical to the slice thickness. Additional details on case selection, dataset splitting, external dataset use, and preprocessing are provided in Supplementary Note 1.

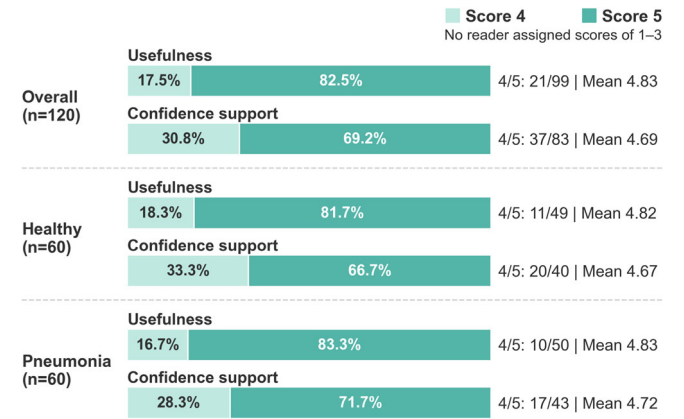


Fig. 7 | Radiologist assessment of synthesized 1 mm CT as an auxiliary view. Usefulness and confidence-support scores are shown for all cases and by health status. Synthesized 1 mm CT was reviewed alongside original 5 mm CT by two radiologists who were blinded to health status. Bars show the distribution of scores 4 and 5 on 5-point scales, with percentages shown within each bar. Values to the right of each bar indicate the number of ratings with score 4 and score 5, followed by the mean score. Overall included 60 cases evaluated by two radiologists, corresponding to 120 reader-case ratings. Healthy and pneumonia subgroups each included 30 cases evaluated by two radiologists, corresponding to 60 reader-case ratings per subgroup. Only scores of 4 and 5 are shown because no reader assigned scores of 1 to 3 for either usefulness or confidence support.

Overall model architecture and training

BasicCSS was designed as a reference model for real-paired CT slice synthesis. It uses a thick-slice CT volume as input and generates a synthesized 1 mm CT volume. The same architecture was used for the 3 mm to 1 mm and 5 mm to 1 mm synthesis tasks, with ratio-specific settings for different axial expansion factors. As shown in Fig. 8a, BasicCSS consists of a volumetric encoder, a slice expansion module, a block-wise decoder, a cross-view refinement module, and a reconstruction tail. The slice expansion module expands the feature volume along the axial direction to the target 1 mm resolution. The block-wise decoder refines local axial feature blocks anchored by adjacent thick-slice positions. The cross-view refinement module further refines sagittal and coronal continuity before final reconstruction, as illustrated in Fig. 8b. The 2D and 3D Swin-Conv blocks used in the encoder and decoder are shown in Fig. 8c.

As part of preprocessing, CT intensities in all input volumes were clipped to [−1024, 2048] HU. The intensity-clipped volumes were then linearly scaled to [0, 1]. The model was optimized using L1 reconstruction loss between the synthesized 1 mm CT volume and the matched real 1 mm CT reference. Optimization used AdamW^{45,46} with a learning rate of 0.0001 and a batch size of 1. During training, input volumes of 8 × 128 × 128 voxels

Table 2 | Radiologist’s assessment of the synthesized 1 mm CT

	Overall (n = 60)		Healthy (n = 30)		Pneumonia (n = 30)	
	Reader 1	Reader 2	Reader 1	Reader 2	Reader 1	Reader 2
Added information, n (%)						
Yes	60 (100.0)	60 (100.0)	30 (100.0)	30 (100.0)	30 (100.0)	30 (100.0)
No	0	0	0	0	0	0
Usefulness, n (%)						
5, very useful	51 (85.0)	48 (80.0)	23 (76.7)	26 (86.7)	28 (93.3)	22 (73.3)
4, useful	9 (15.0)	12 (20.0)	7 (23.3)	4 (13.3)	2 (6.7)	8 (26.7)
1-3, not useful to moderately useful	0	0	0	0	0	0
Confidence support, n (%)						
5, strong support	42 (70.0)	41 (68.3)	18 (60.0)	22 (73.3)	24 (80.0)	19 (63.3)
4, clear support	18 (30.0)	19 (31.7)	12 (40.0)	8 (26.7)	6 (20.0)	11 (36.7)
1-3, no support to moderate support	0	0	0	0	0	0

Values are shown as n (%). Overall includes 60 EC-CT cases evaluated independently by each reader. Healthy and pneumonia subgroups included 30 cases each. Added information was rated as yes or no. Usefulness and confidence support were scored on 5-point scales, with higher scores indicating greater usefulness or stronger support for interpretation confidence. Scores of 1 to 3 were combined because no ratings below 4 were assigned. For usefulness, scores of 1, 2, and 3 indicated not useful, slightly useful, and moderately useful, respectively. For confidence support, scores of 1, 2, and 3 indicated no support, slight support, and moderate support, respectively.

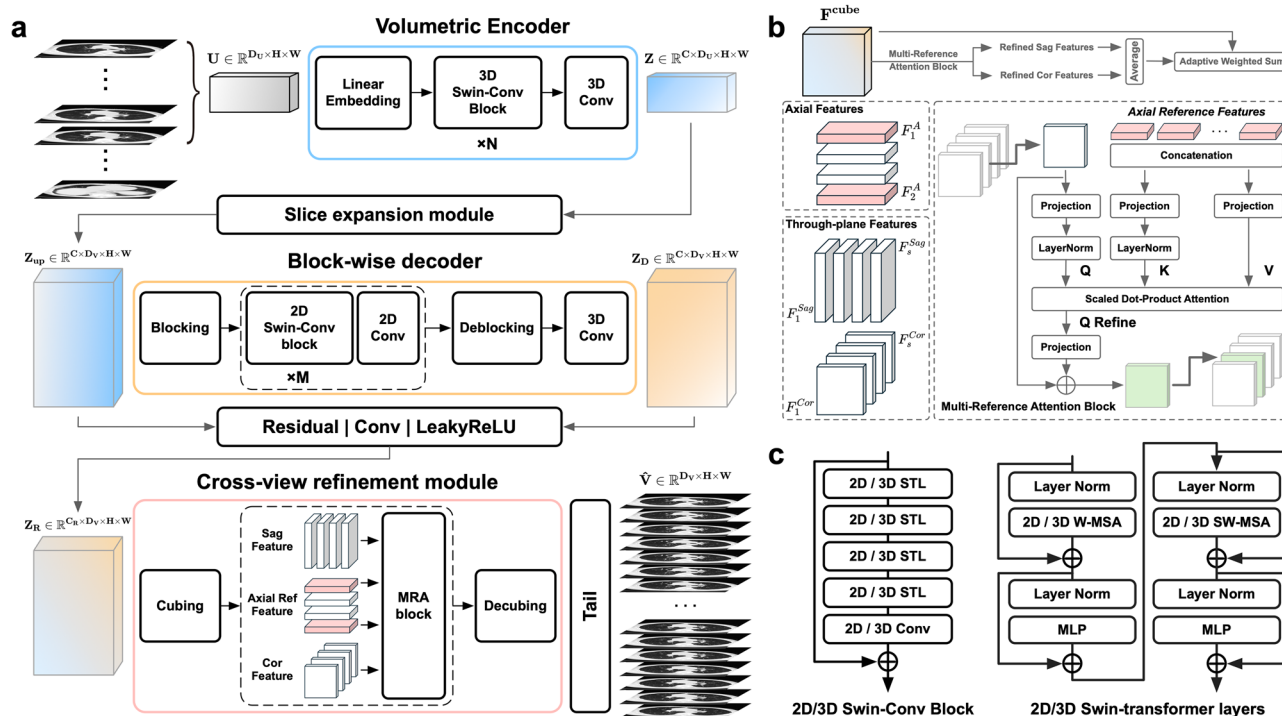


Fig. 8 | BasicCSS model architecture. **a** Overall architecture of BasicCSS. A thick-slice CT volume is processed by a volumetric encoder to extract three-dimensional features. The slice expansion module expands the feature volume along the axial direction to the target thin-slice resolution. The expanded features are refined by a block-wise decoder using blocking and deblocking operations, followed by residual aggregation. The cross-view refinement module further refines the reconstruction-ready feature volume before the reconstruction tail maps it to the image domain and generates the synthesized 1 mm CT volume. **b** Cross-view refinement module and multi-reference attention block. The feature volume is partitioned into local

volumetric regions. Within each region, axial reference features are used to refine sagittal and coronal feature representations through multi-reference attention. The refined sagittal and coronal features are then combined and mapped back to the volumetric feature space. **c** Structure of the 2D and 3D Swin-Conv block and the corresponding Swin-Transformer layers used in the encoder and decoder. Cor coronal, CRM cross-view refinement module, MLP multilayer perceptron, MRA multi-reference attention, Sag sagittal, STL Swin-transformer layer, SW-MSA shifted-window multi-head self-attention, W-MSA window-based multi-head self-attention.

were used, with random cropping and flipping applied for data augmentation. During inference, thick-slice CT volumes were synthesized using a sliding-window strategy. Detailed module definitions and additional training and inference details are provided in Supplementary Note 2.

Ablation study. To examine the contributions of the block-wise decoder and cross-view refinement module, BasicCSS was compared with ablated variants under the same data splits, preprocessing, loss function, and training protocol. Quantitative performance was evaluated using PSNR and three-plane SSIM on the MSS-CT 3 mm and MSS-CT 5 mm internal test sets. Detailed model variants and convergence analyses are provided in Supplementary Note 3.

External-center adaptation analysis. To assess whether MSS-CT training could support external-center adaptation, we compared three BasicCSS settings: training from scratch on external-center training cases, direct evaluation of the MSS-CT-trained model without external-center training, and fine-tuning of the MSS-CT-trained model using external-center training cases. The analysis was performed on both RPLHR-CT and EC-CT, and all settings were evaluated on held-out external test sets using PSNR and three-plane SSIM. Complete data splits and training settings are provided in Supplementary Note 4.

Comparator methods

Trilinear interpolation was used as the representative conventional interpolation method. Deep learning-based comparator methods were selected to cover the main methodological formulations used in prior CT slice synthesis studies, including reformatted-view super-resolution, volume super-resolution, and slice-wise interpolation^{12–21}. To broaden the

benchmark, representative natural image super-resolution^{37–39} and video frame interpolation models^{40–42} were also adapted to the CT slice synthesis setting. For methods with publicly available code, official implementations were adapted to the present task; otherwise, methods were reimplemented according to the original publications. To reduce the influence of model size on method comparison, the numbers of learnable parameters was kept within comparable ranges where possible. All comparator methods were evaluated using the same data splits, preprocessing, training protocol, and test conditions. Full details of the evaluated methods and their implementation settings are provided in Supplementary Note 5.

Synthesis performance evaluation

Synthesis performance was evaluated for trilinear interpolation, BasicCSS, and the comparator CSS methods on the MSS-CT 3 mm and MSS-CT 5 mm internal test sets. For the 5 mm to 1 mm synthesis task, external evaluation was also performed on RPLHR-CT and EC-CT. Models trained on MSS-CT 5 mm were directly tested on the two external datasets without fine-tuning. Each synthesized or resampled 1 mm CT series was compared with the matched real 1 mm CT reference. Performance was assessed using PSNR calculated over the whole CT volume and SSIM calculated separately in the axial, coronal, and sagittal planes. Representative cases from all four test datasets were selected for qualitative comparison. Detailed definitions for PSNR and SSIM are provided in Supplementary Note 6.

Real-paired and pseudo-paired training comparison

To assess the effect of training-pair realism, real-paired training was compared with two pseudo-paired training strategies on the MSS-CT 3 mm and MSS-CT 5 mm datasets. The comparison included BasicCSS and representative methods from the three main CSS paradigms, including ACVT²¹

for reformatted-view super-resolution, ResVox¹³ for volume super-resolution, and CTHNet¹⁸ for slice-wise interpolation. Two recent pseudo-paired methods, I3Net¹⁹ and SFCLI-Net²⁰, were also included.

In the real-paired setting, the model input was the clinically reconstructed thick-slice CT series. In the sampled pseudo-paired setting, pseudo thick-slice inputs were generated by sampling slices from real 1 mm CT at fixed axial intervals corresponding to the target synthesis ratio. In the degraded pseudo-paired setting, pseudo thick-slice inputs were generated from real 1 mm CT by axial degradation to simulate thick-slice inputs. Apart from input construction, all settings used the same data splits, model architecture, loss function, optimizer, training schedule, and data augmentation. After training, all models were evaluated under real-paired test conditions using real thick-slice CT as input and matched real 1 mm CT as the reference. Additional implementation details are provided in Supplementary Note 7.

Segmentation-derived structural consistency analysis

Segmentation-derived structural consistency was evaluated to assess whether synthesized 1 mm CT produced downstream anatomical segmentation outputs consistent with those derived from real 1 mm CT. This analysis was performed on the complete MSS-CT 3 mm internal test set, the complete MSS-CT 5 mm internal test set, the predefined 100-case RPLHR-CT test set, and 100 randomly selected EC-CT cases. TotalSegmentator (v2.11)²⁵ was applied with default settings to matched real 1 mm CT, trilinear interpolation-resampled 1 mm CT (Tri-CT), and BasicCSS-generated 1 mm CT (CSS-CT). Masks derived from real 1 mm CT were used as reference segmentations, and Tri-CT- and CSS-CT-derived masks were compared with the reference masks using the Dice similarity coefficient. Additional details on the evaluated structures and Dice calculation are provided in Supplementary Note 8.

Radiomic feature consistency analysis

Radiomic feature consistency was evaluated to assess whether synthesized 1 mm CT generated quantitative image features consistent with those extracted from real 1 mm CT. This analysis was performed on the same four datasets used for the segmentation-derived structural consistency analysis. For each case, a lung ROI defined on real 1 mm CT was used for feature extraction from real 1 mm CT, thick-slice CT, trilinear interpolation-resampled 1 mm CT (Tri-CT), and BasicCSS-generated 1 mm CT (CSS-CT). Radiomic features extracted from real 1 mm CT were used as the reference, and the concordance correlation coefficient (CCC) was calculated for each feature across cases. A feature was considered reproducible if its CCC was at least 0.85. Additional details on ROI sampling, feature extraction, and CCC calculation are provided in Supplementary Note 8.

Reader study

A reader study was performed to assess whether BasicCSS-generated 1 mm CT could serve as an auxiliary view when only original 5 mm CT was available. The assessment focused on diagnostic information, visible anatomical detail, usefulness for image review, and support for interpretation confidence rather than visual appearance alone, in line with established CT image-quality criteria⁴³. Sixty EC-CT cases were randomly selected, including 30 healthy cases and 30 patients with pneumonia. Two radiologists with 10 and 3 years of experience independently reviewed original 5 mm CT and BasicCSS-generated 1 mm CT side by side. The real 1 mm CT was not shown because the study was designed to evaluate the synthesized image as an auxiliary view rather than as a replacement for real thin-slice CT. Readers were aware of which series was original and which was synthesized, but were blinded to health status and quantitative evaluation results. Readers evaluated added information, usefulness, and confidence support. Added information was rated as yes or no, and usefulness and confidence support were scored on 5-point scales. Additional details on the reading procedure, rating criteria, and definition of added information are provided in Supplementary Note 9.

Statistical analysis

Continuous variables are reported as mean $\pm$ standard deviation unless otherwise specified. For box plots, the center line indicates the median, and the box indicates the interquartile range. For quantitative bar plots, bars show mean values and error bars indicate standard deviation. For stacked rating bars in the reader study, bars show score distributions. Paired comparisons were performed using two-sided paired Wilcoxon signed-rank tests because the compared methods, training settings, or image types were evaluated on the same cases or the same radiomic features. When multiple pairwise comparisons were performed, *p* values were adjusted using Bonferroni correction within the corresponding comparison family.

For synthesis performance evaluation, BasicCSS was compared with each comparator method using case-level PSNR and three-plane SSIM. For the real-paired and pseudo-paired training comparison, case-level PSNR and three-plane SSIM were compared between training-pair settings within each synthesis task, method, and metric. In this analysis, real-paired training was compared separately with each pseudo-paired training strategy.

For segmentation-derived structural consistency analysis, Dice values were calculated for each case and anatomical structure. CSS-CT and Tri-CT were compared for each anatomical structure within each dataset, using the same real 1 mm CT-derived reference masks. Dice values are reported as percentages in the figures and tables.

For radiomic feature consistency analysis, the concordance correlation coefficient (CCC) was calculated for each radiomic feature across cases, comparing thick-slice CT, Tri-CT, or CSS-CT with the matched real 1 mm CT reference. CCC distributions were compared at the feature level, with pairing defined by the same radiomic feature across image types. The proportion of reproducible features, defined as features with CCC $\geq$ 0.85, was summarized descriptively.

Reader study results were summarized descriptively. Added information was reported as *n* (%). Usefulness and confidence-support scores were summarized by score distribution and mean score, using all ratings from the two readers. Subgroup summaries were reported for healthy and pneumonia cases.

Statistical significance was indicated as * for adjusted *p* < 0.05, ** for adjusted *p* < 0.01, and *** for adjusted *p* < 0.001. All statistical analyses were performed in Python 3.9.

Data availability

The MSS-CT dataset is available at <https://github.com/smilenaxx/MSS-CT>. Only de-identified NIfTI image volumes are released. Before conversion to NIfTI format, all DICOM fields containing direct or indirect personal identifiers were removed. Each case was assigned a random study identifier that was not derived from any clinical identifier, and no file linking the released identifiers to the original hospital records was retained in the public dataset. The released NIfTI files contain image volumes only, and their headers were checked to ensure that they did not contain personal identifiers or institution-specific information. The case assignments are provided with the released dataset. The RPLHR-CT dataset used in this study is available through the access link provided in the original publication at <https://zenodo.org/records/17239183>. The EC-CT dataset is not publicly available because of institutional regulations and patient privacy restrictions. Researchers may request access to de-identified clinical data by contacting the corresponding author, Y.W. Access will be subject to institutional review board approval and execution of a data-use agreement, and will be limited to noncommercial academic research. Requests will be reviewed within 4 weeks.

Code availability

The BasicCSS code is available at <https://github.com/smilenaxx/MSS-CT>. All experiments were implemented in Python 3.9.23 using PyTorch 2.2 and performed on NVIDIA RTX A6000 GPUs with 48 GB memory.

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Author contributions

P.Y., X.Y., Y.W., and S.Q. designed the study. P.Y., H.Z. and S.Z. led method development, wrote the code, and performed the experiments. P.Y., X.Y., C.P., and Y.W. collected the data. X.Y. and M.X. performed the reader study. P.Y. and X.Y. drafted the manuscript. H.Z., S.Z., X.C., D.W., W.Q., S.H., B.T., Y.W., and S.Q. critically revised the manuscript. X.Y., Y.W., and S.Q. are co-corresponding authors. All authors read and approved the final manuscript.

Competing interests

P.Y. is an intern at Infervision Medical Technology Co. Ltd, X.C. and D.W. are employed by Infervision Medical Technology Co., Ltd. All other authors declare no competing interests.

Additional information

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